

Two obstructions to an infinite common ℓ_p spectrum

Partial progress on Problem 1 of arXiv:0206181

Candidate solution packet

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Abstract

Tokarev asked whether a finite-equivalence class X^f can have infinitely many exponents p for which every representative contains ℓ_p . We do not settle the problem, but prove two useful reductions. First, no finite exponent can belong to the divisibility index of a positive example. Second, the most direct countable-sum construction fails sharply: a sum containing prescribed ℓ_{q_n} coordinates is finitely equivalent to an ℓ_2 -sum of finite-dimensional blocks containing none of those ℓ_{q_n} spaces.

1. Exact problem and notation

For a Banach space Y , the source writes

$$T(Y) = \{p : \ell_p \text{ is isomorphic to a subspace of } Y\}.$$

If X^f denotes the class of all Banach spaces finitely equivalent to X , Problem 1 asks verbatim:

“Whether there exists such a class X^f that $\cap\{T(Y) : Y \in X^f\}$ is infinite?”

$$T(X) = \{p : \ell_p \text{ is isomorphic to a subspace of } X\}.$$

Let $Y, Z \in X^f$. Let both Y and Z be separable and stable. Thus, $T(Y) \neq \emptyset$ and $T(Z) \neq \emptyset$. Assume that $T(Y) \cap T(Z) = \emptyset$. Similarly to the proof of the theorem 2 may be constructed a space $W \in X^f$ that has the Tsirelson property. However this conflicts with the superstability of X^f . Hence $\cap\{T(Y) : Y \in X^f\} \neq \emptyset$. $\square$

REMARK 7. *There are classes X^f such that the intersection $\cap\{T(Y) : Y \in X^f\}$ consists exactly of n points.*

Indeed, let $X = \ell_{q_1} \oplus \ell_{q_2} \oplus \dots \oplus \ell_{q_n}$, where $2 < q_1 < q_2 < \dots < q_n$. Clearly,

$$\cap\{T(Y) : Y \in X^f\} = \{q_1, q_2, \dots, q_n\}.$$

PROBLEM 1. *Whether there exists such a class X^f that $\cap\{T(Y) : Y \in X^f\}$ is infinite?*

In what follows will be needed a definition.

DEFINITION 13. *(Cf. [5]) Let X be a Banach space, $(x_n) \subset X$ be a nontrivial normed sequence of elements of X (i.e. (x_n) contains no Cauchy subsequences); D be an ultrafilter over $\mathbb{N}$. For a finite sequence $(a \cdot)^n \subset \mathbb{R}^n$ let*

The source's divisibility index is

$$\text{Index}(X^f) = \{r \in [1, \infty] : Y \in X^f \implies \ell_r(Y) \in X^f\}.$$

It also proves a saturation theorem. Every r -divisible class contains a representative Y_r such that each infinite-dimensional subspace of Y_r contains an isomorphic copy of ℓ_r .

2. A finite-divisibility obstruction

Proposition 1 (Finite divisibility obstruction). *If $r \in \text{Index}(X^f)$ for some $1 \leq r < \infty$, then*

$$\left| \bigcap_{Y \in X^f} T(Y) \right| \leq 1.$$

In particular, the divisibility index of any positive answer to Problem 1 contains no finite exponent.

Proof. By the saturation theorem quoted above, there is an ℓ_r -saturated $Y_r \in X^f$. If $\ell_q \hookrightarrow Y_r$, then the infinite-dimensional subspace ℓ_q must itself contain ℓ_r . Classical total incomparability of the sequence spaces ℓ_q and ℓ_r for $q \neq r$ makes this impossible. Thus $T(Y_r) \subseteq \{r\}$ and the common intersection is contained in $\{r\}$. $\square$

This removes every self-similar construction such as a repeated ℓ_r -sum: once reindexing gives $\ell_r(X) \simeq_f X$, the class is r -divisible and Proposition 1 applies.

3. Why the obvious countable direct sum fails

The next result records a stronger, explicit failure. Fix distinct exponents $q_n \in (1, \infty) \setminus \{2\}$. Let

$$X = \left(\bigoplus_{(n,k) \in \mathbb{N}^2} \ell_{q_n} \right)_2,$$

$$Z = \left(\bigoplus_{(n,k,m) \in \mathbb{N}^3} \ell_{q_n}^m \right)_2.$$

Every finite block type is repeated infinitely often in Z .

Proposition 2 (Finite-block erasure). *The spaces X and Z are finitely equivalent. The space X contains an isometric copy of every ℓ_{q_n} , whereas Z contains no isomorphic copy of any ℓ_{q_n} .*

Proof. Every finite-dimensional subspace of X is, up to arbitrarily small distortion, supported on finitely many outer coordinates. Its projection into each of those ℓ_{q_n} coordinates is finite dimensional and can, again with arbitrarily small distortion, be moved into some coordinate copy of $\ell_{q_n}^m$ in Z . Distinct outer coordinates are sent to distinct blocks, so the outer ℓ_2 norm is preserved. Hence $X <_f Z$.

Conversely, a finite-dimensional subspace of Z is approximated by finitely many of its finite-dimensional blocks. Each block $\ell_{q_n}^m$ embeds isometrically into a fresh copy of ℓ_{q_n} in X ; the infinitely many copies indexed by k let us keep different blocks in different outer coordinates. Thus $Z <_f X$.

The coordinate inclusion shows $\ell_{q_n} \hookrightarrow X$. To see that none embeds in Z , use the finite-dimensional decomposition of Z into its displayed finite blocks. Every normalized weakly null sequence in an ℓ_2 -sum of finite-dimensional blocks has, by the usual gliding-hump argument, a subsequence arbitrarily close to a normalized block sequence. Disjoint normalized blocks are isometrically equivalent to the unit vector basis of ℓ_2 .

If $J : \ell_{q_n} \rightarrow Z$ were an isomorphic embedding, then (Je_j) would be a seminormalized weakly null sequence, because $1 < q_n < \infty$. A subsequence would therefore be equivalent to the ℓ_2 basis. It

is also equivalent to a subsequence of the symmetric ℓ_{q_n} basis, hence to the ℓ_{q_n} basis itself. This is impossible when $q_n \neq 2$, as the norm of the sum of the first N unit vectors grows as N^{1/q_n} rather than $N^{1/2}$. $\square$

Thus local finite-dimensional data can retain every prescribed $\ell_{q_n}^m$ while erasing every corresponding infinite-dimensional ℓ_{q_n} copy. A full positive solution must force infinitely many infinite types by a mechanism substantially stronger than repeated direct sums or finite representability alone.

4. What remains

The reductions leave a sharp residual problem:

Construct a finite-equivalence class with no finite divisibility exponent in which infinitely many distinct ℓ_p types are unavoidable in every representative, or prove that no such class exists.

An attempted isolated-exponent route would require a realization theorem turning isolated local ℓ_p data into an actual subspace in every stable representative. Complemented copies in ultrapowers do not descend in this generality, so that route presently has a genuine gap.

Bounded searches through August 11, 2026 found no later source stating an answer to Tokarev's problem. The two propositions above are elementary but appear not to be recorded in the source. Priority is not asserted, and specialist review is recommended.

References

- [1] E. V. Tokarev, *On Banach spaces with the Tsirelson property*, arXiv:math/0206181 (2002), Problem 1.
- [2] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I: Sequence Spaces*, Springer, 1977.